Quantization and language: a focused AfriXNLI analysis

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Exploratory revision for methodological review. Not preregistered or peer reviewed.

One question

For the released Qwen2.5-1.5B-Instruct GGUF artifacts, does the Q4_K_M-minus-Q8_0 accuracy change differ between Yoruba and English on AfriXNLI?

I have narrowed the earlier multi-provider benchmark to one model, one task, two languages and two released configurations. This note re-analyses existing observations; it contains no new model inference.

Input language	Q8_0	Q4_K_M
English	300 questions	Same 300 questions
Yoruba	300 parallel translations	Same 300 translations

Design and primary outcome

The primary contrast is the Yoruba Q4-minus-Q8 accuracy change minus the English change. Negative values mean a greater measured Yoruba loss. Each parallel question carries all four outcomes: 300 paired clusters, not 1,200 independent questions.

Both artifacts share a pinned official GGUF release. The recorded workspace, runtime binary, CPU-only two-thread settings, 2,048-token context, English instructions/labels, greedy decoding, seed, 128-token answer budget and strict scorer are matched. For each item I also check that request sizes and reported prompt-token counts match between configurations.

I independently reparse and rescore the responses, reject duplicate successful answers or missing cells, and preserve request failures. I resample parallel IDs jointly for 10,000 bootstrap replicates, seed 20261102, and report nominal 95% percentile intervals. These intervals are conditional on the recorded runs.

Boundaries

The scope was narrowed after results were inspected. The original conditions ran sequentially on different dates on a shared host. The full upstream conversion/calibration chain is not recorded. Q8_0 is compressed; the recipes differ as well as nominal precision. This is an exploratory comparison of released configurations, not a pure causal bit-width experiment.

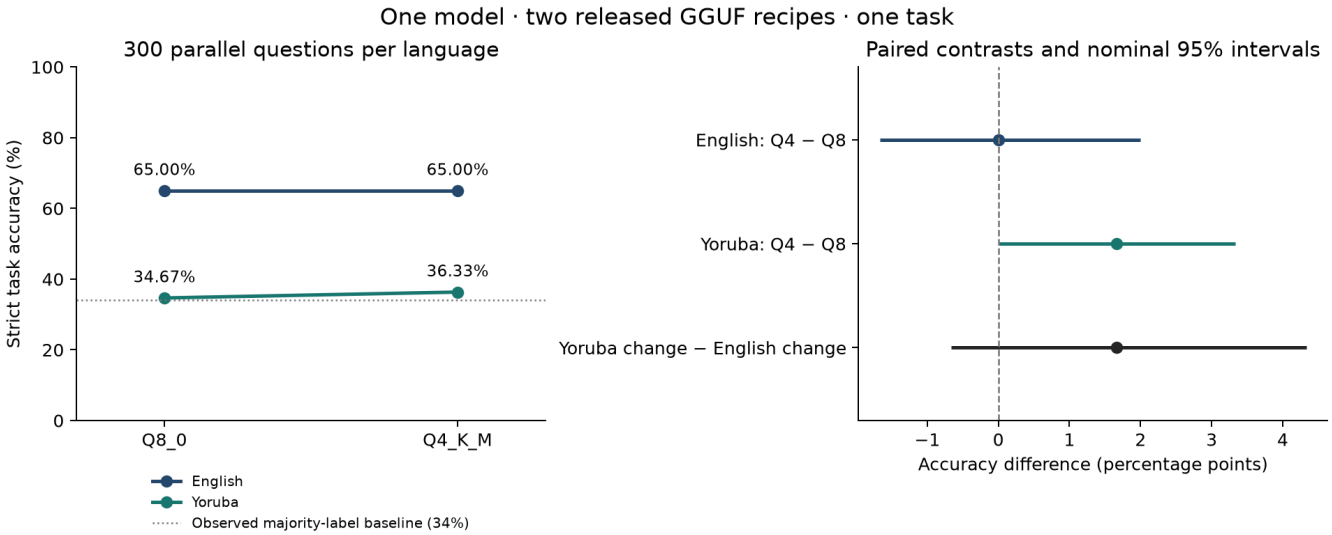
I exclude other tasks, languages, model families, GPU/API providers, translation pipelines and timing repeats. I make no serving-cost, latency, phone or production claim.

Measured result

Primary contrast: +1.67 pp [-0.67, +4.33]. The interval spans zero. These observations do not establish a greater Yoruba compression penalty, equivalence, or a general effect of bit width.

Language	Configuration	Correct / 300	Accuracy
eng	Q8_0	195	65.00%
eng	Q4_K_M	195	65.00%
yor	Q8_0	104	34.67%
yor	Q4_K_M	109	36.33%

Paired contrast	Estimate [nominal 95% interval]
English: Q4 minus Q8	+0.00 pp [-1.67, +2.00]
Yoruba: Q4 minus Q8	+1.67 pp [+0.00, +3.33]
Yoruba change minus English change	+1.67 pp [-0.67, +4.33]



The bootstrap resamples all four outcomes together. Equal English accuracy still hides answer changes: four questions are correct only in Q4 and four only in Q8. The corresponding Yoruba counts are six and one. These are question-level observations from single recorded condition runs.

All 1,200 final responses are parseable, and none stops at the output token cap. One English Q8 request failed before a successful retry; that attempt remains in the raw log. Formatting failure and output caps therefore do not explain the four accuracy values.

Diagnostics and remaining controls

Class bias limits interpretation

Language	Configuration	Neutral recall
eng	Q8_0	0.98%
eng	Q4_K_M	4.90%
yor	Q8_0	0.00%
yor	Q4_K_M	0.00%

Neither Yoruba configuration ever predicts neutral. English neutral recall is poor too. Gold counts are identical across languages: 99 entailment, 102 neutral and 99 contradiction; the observed majority-label baseline is 34%. Yoruba predictions heavily favour entailment. I have not established whether the cause is model capability, prompting, translations or another source.

Reported prompt-token ranges are 72–124 in English and 92–281 in Yoruba. The study uses English instructions and labels with translated premises/hypotheses. I did not change the prompt or relax the parser after observing the failures.

What still needs a fresh design

A shared host and sequential collection leave execution-order and contention confounding. Question bootstrap intervals do not capture this, run-to-run variation, translation error or post-hoc selection. The official artifact revision is pinned, but the full common unquantized-weight/conversion lineage needs recording for a compression-process claim. Independent native review of these benchmark translations has not been completed; the agent-safety volunteers reviewed a different dataset.

I have reserved the unused 300-question complement of the same 600-row benchmark. No confirmation inference has happened. The proposal is to freeze a bounded execution plan on a reserved host, document common conversion lineage, balance configuration order across question blocks, preserve all failures and preselect a repeat-agreement check. Block order and repeated-subset IDs are not frozen yet. This is a within-benchmark holdout, not an independent corpus.

What I would like reviewed

Is the language-by-configuration contrast an appropriate focused target? Given the severe class bias, should a separate development phase establish a suitable prompt/model before any confirmatory run?

Evidence and reproduction

Protocol, immutable input hashes, paired outcomes, diagnostics and analysis code.

From the repository root: install the project with its dev and review extras; run python -m bench.data, python -m bench.focused_nli, python -m bench.focused_review and pytest -q. Data preparation retrieves the pinned benchmark; the analysis makes no model requests. The original run files and broad protocol remain unchanged.